BobSim

TransientEval Lateral Transient Response

Comprehensive Open-Loop Characterization

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BobDyn

– One-period sinusoidal input

– Continuous sinusoidal response

– Frequency response computed from fixed-amplitude, single-direction sine sweep

– Steering shown in degrees for readability

TransientEval Metrics Summary

Time Domain — Step Response

STEP RESPONSE

Peak a_y	5.58	m/s²
Steady-State a_y	5.10	m/s²
Steady-State Gain	58.48	(m/s²)/rad
Overshoot (a_y)	9.3	%
Rise Time (10–90%)	0.11	s
Settling Time	1.23	s
Yaw Overshoot	14.1	%

TransientEval Metrics Summary

Frequency Domain — Sustained Sine

FREQUENCY RESPONSE — CORE

DC Gain (a_y/δ_H)	55.42	(m/s²)/rad
DC Gain (r/δ_H)	2.77	(rad/s)/rad
Peak Gain (a_y)	55.79	(m/s²)/rad
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	2.00	Hz
Phase @ 1 Hz (a_y)	-13.6	deg
Phase @ 1 Hz (r)	-10.5	deg
Lag @ 1 Hz (a_y)	0.038	s
Lag @ 1 Hz (r)	0.029	s

DYNAMIC CHARACTER

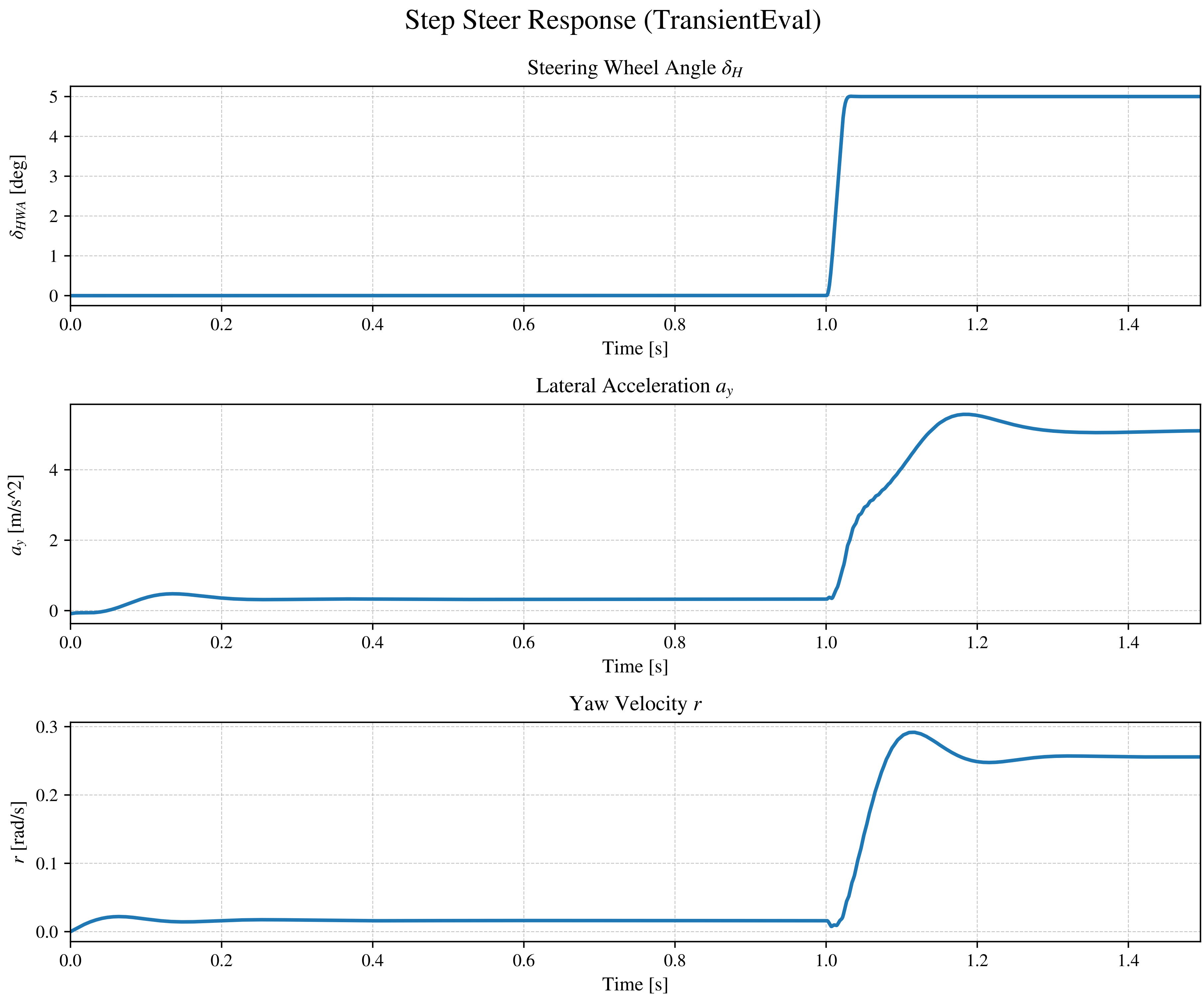
Gain Slope (a_y)	-0.04	dB/dec
Gain Slope (r)	0.51	dB/dec
Phase Slope (a_y)	41.02	deg/Hz
Phase Slope (r)	26.89	deg/Hz
Phase = -45° (a_y)	2.00	Hz
Phase = -45° (r)	2.00	Hz

RESPONSE COUPLING

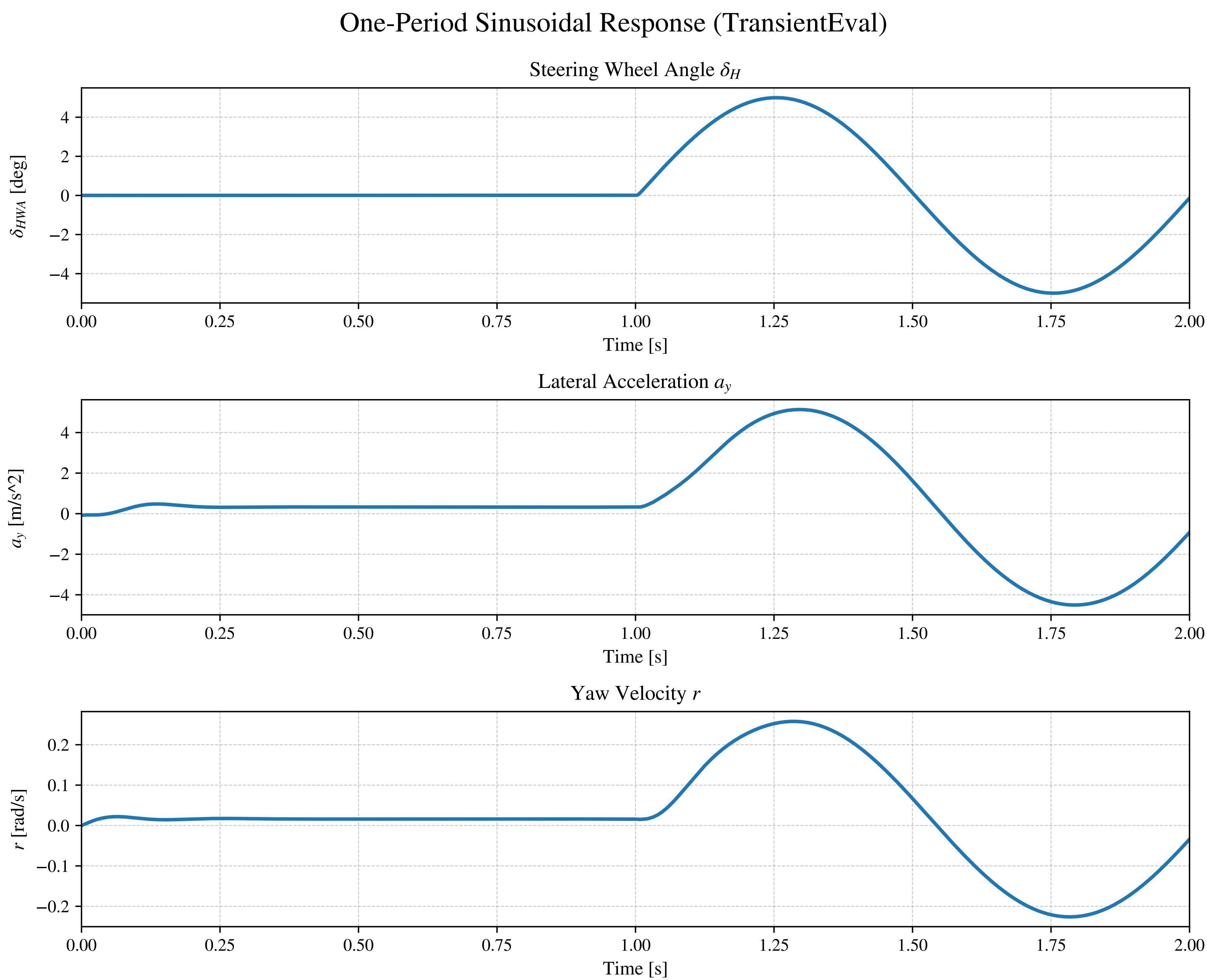
Lag: $\delta_H \rightarrow a_y$	0.038	s
Lag: $\delta_H \rightarrow r$	0.029	s
Lag: $r \rightarrow a_y$	0.009	s
r/a_y Ratio	0.050	(rad/s)/(m/s²)

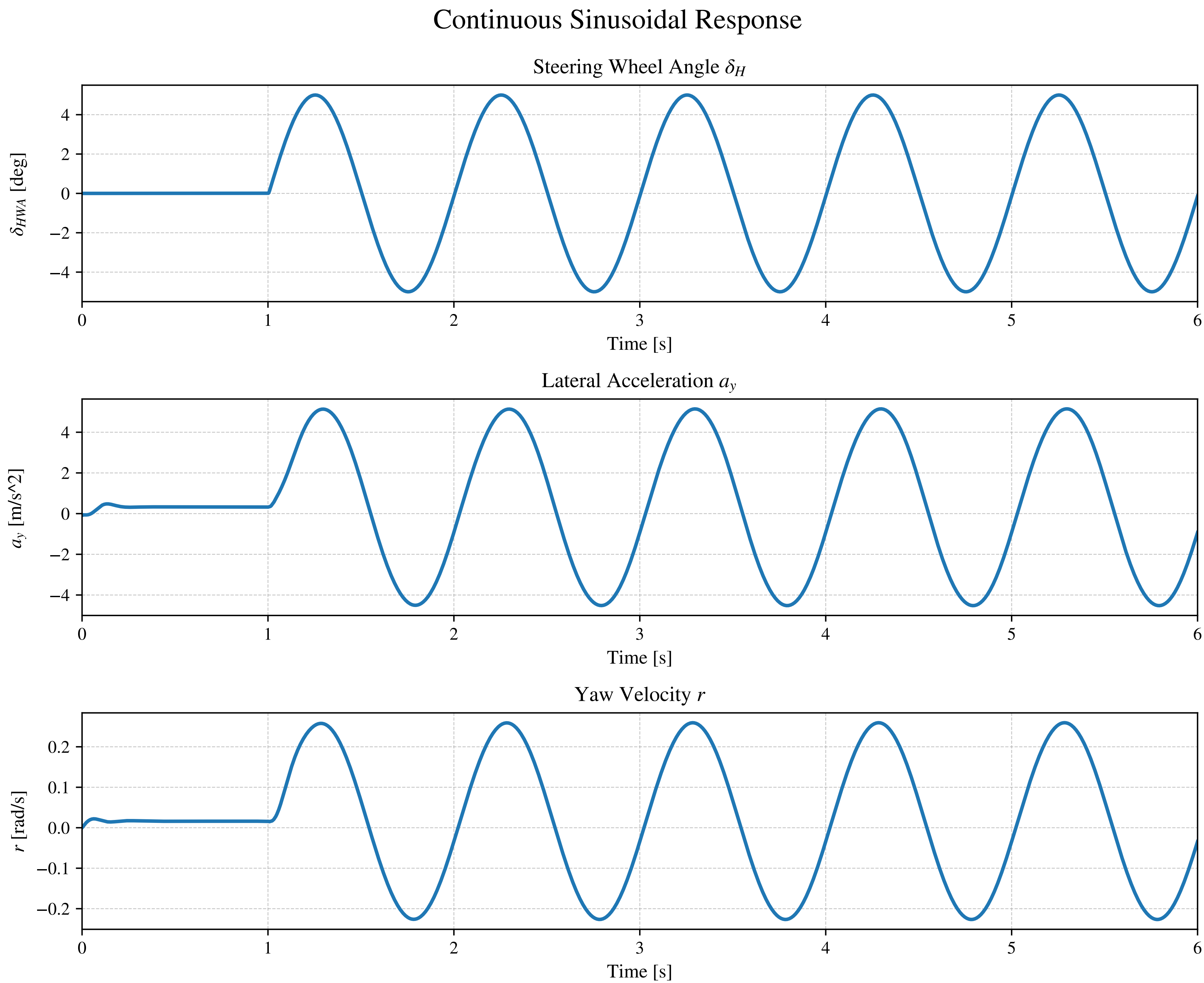
QUALITY / VALIDITY

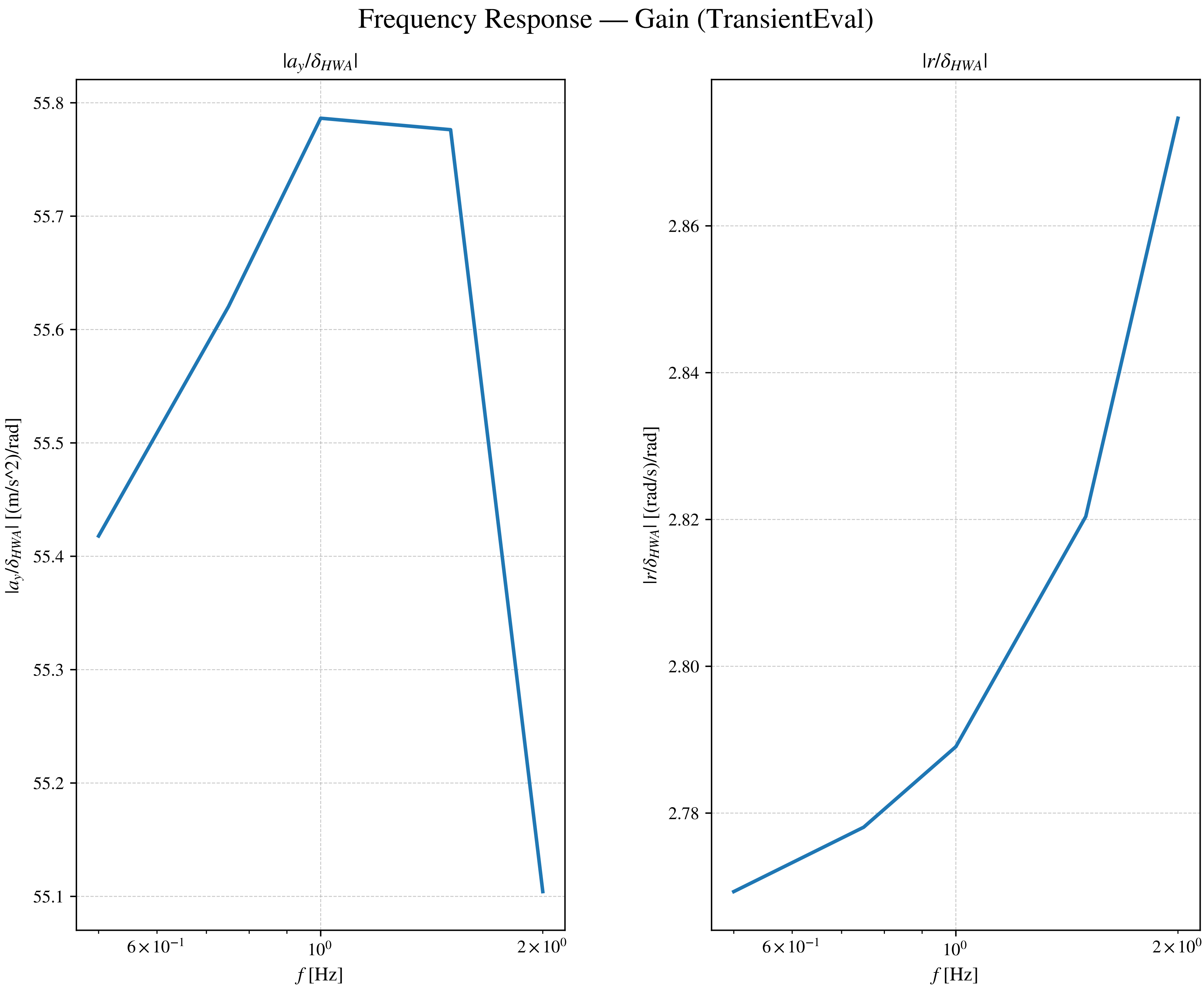
a_y Fit Error	1.15e-02	-
Yaw Fit Error	4.17e-03	-
Gain Variation	1.2	%



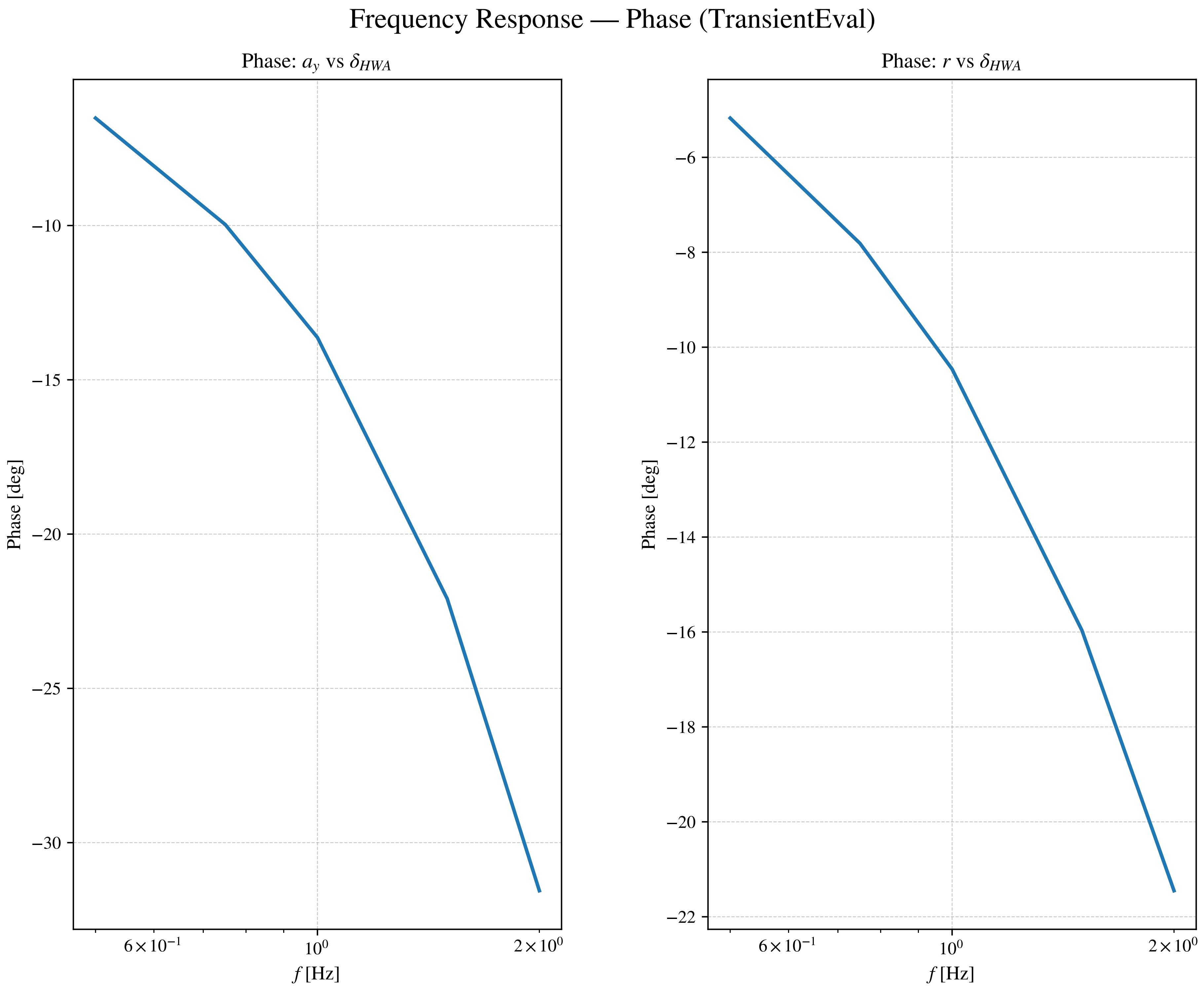
Step Steer Response (TransientEval)
Steering Wheel Angle δ_H
δ_{HWA} [deg]
0
1
2
3
4
5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time [s]
Lateral Acceleration a_y
a_y [m/s^2]
0
2
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time [s]
Yaw Velocity r
r [rad/s]
0.0
0.1
0.2
0.3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time [s]


One-Period Sinusoidal Response (TransientEval)
Steering Wheel Angle δ_H
δ_{HWA} [deg]
Time [s]
Lateral Acceleration a_y
a_y [m/s^2]
Time [s]
Yaw Velocity r
r [rad/s]
Time [s]


Continuous Sinusoidal Response
Steering Wheel Angle δ_H
δ_{HWA} [deg]
4
2
0
−2
−4
0
1
2
3
4
5
6
Time [s]
Lateral Acceleration a_y
a_y [m/s^2]
4
2
0
−2
−4
0
1
2
3
4
5
6
Time [s]
Yaw Velocity r
r [rad/s]
0.2
0.1
0.0
−0.1
−0.2
0
1
2
3
4
5
6
Time [s]



Frequency Response — Gain (TransientEval)
$|a_y/\delta_{HWA}|$
$|r/\delta_{HWA}|$
55.8
55.7
55.6
55.5
55.4
55.3
55.2
55.1
$|a_y/\delta_{HWA}|$ [(m/s^2)/rad]
2.86
2.84
2.82
2.80
2.78
$|r/\delta_{HWA}|$ [(rad/s)/rad]
6×10^{-1}
10^0
2×10^0
f [Hz]



Frequency Response — Phase (TransientEval)
Phase: a_y vs δ_{HWA}
Phase [deg]
−10
−15
−20
−25
−30
6×10^{-1}
10^0
2×10^0
f [Hz]
Phase: r vs δ_{HWA}
Phase [deg]
−6
−8
−10
−12
−14
−16
−18
−20
−22
6×10^{-1}
10^0
2×10^0
f [Hz]